

Unsupervised Detection of Political Homophily Through Positive Mention Networks on X During the 2024 U.S. Presidential Election

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June 10, 2026

Abstract

Social media platforms like X (formerly Twitter) have transformed the way people engage with politics, giving millions of users the ability to share opinions, follow political figures, and interact with others in real time. However, this openness comes with a growing concern that rather than exposing users to a diverse range of viewpoints, social media may be reinforcing existing beliefs by encouraging people to interact almost exclusively with others who think like them. This phenomenon, known as an echo chamber, has been linked to increased political polarisation, the spread of misinformation, reduced exposure to opposing perspectives, and a general decrease in constructive political dialogue in democratic societies. [14, 9]

In this study, we investigate whether users who positively mention each other on X during the 2024 U.S. Presidential Election tend to belong to the same political side. We use a large dataset of election-related tweets and build interaction networks for two time periods (May–July 2024 and October 2024), keeping only mentions that carry a neutral or positive sentiment. We then apply the Louvain community detection algorithm, without providing any political labels in advance, to discover natural user groupings from the interaction data alone. We measure how isolated each community is using the E/I index and validate the political identity of each group by inspecting the most mentioned accounts within it.

Our results show that echo chambers are present and strong throughout both time periods. Surprisingly, they did not grow stronger closer to election day. Most communities were slightly more open in October than in May. Overall, our findings suggest that who users choose to positively engage with on X is a strong indicator of their political ideology, and that echo chambers are a persistent feature of political discourse on social media regardless of how close an election is.

Keywords: Echo chambers, political homophily, Twitter/X, mention networks, community detection, Louvain algorithm, political polarisation, unsupervised learning, 2024 U.S. Presidential Election, temporal network analysis.

1 Introduction

Understanding how echo chambers form and whether they grow stronger over time has been an important area of research in recent years. We reviewed several key papers in this area and found a number of recurring limitations that shaped the design of our own study.

Marques-Neto et al. [16] studied political homophily on Twitter during the 2016 U.S. Presidential Election, using sentiment analysis to group users as Trump or Hillary supporters and measure how much they tended to interact

with like-minded others. While this work laid important groundwork, the data is now over ten years old. Since then, Twitter was acquired and rebranded as X, its recommendation algorithm and content moderation policies was significantly changed. Interaction patterns observed in 2016 cannot be assumed to hold under the very different platform conditions that existed during the 2024 election.

Esteve-Del-Valle [7] compared homophily and political polarisation in Twitter mention networks across two regions/countries (Catalonia and the Netherlands) measuring partisan clustering among Members of Parliament using the E/I index. While informative, both countries use multi-party proportional representation systems, which produce very different network structures compared to the United States. The U.S. operates under a dominant two-party system where political discourse is almost entirely structured around two deeply opposing camps, making comparisons with European multi-party systems difficult to generalise.

Enjolras et al. [6] examined homophily and echo chambers on Twitter during the 2017 Norwegian parliamentary election and found no clear evidence of echo chambers, explaining this by pointing to Norway's generally low level of political polarisation. This raises an important question: would the same result be found in a far more polarised political environment? The 2024 U.S. Presidential Election, contested between Donald Trump and Kamala Harris under conditions of historically high political tension [4], offers exactly such a context where echo chambers are far more likely to emerge and be detectable.

Cota et al. [18] studied echo chambers using 12 million tweets about the impeachment of Brazilian President Dilma Rousseff. Although this paper offered a strong framework, it focused on a single event in a single country, and the authors themselves called for the work to be replicated in other political contexts. Additionally, their method of classifying political leaning relied entirely on manually tagged

hashtags, which the authors acknowledged as a weakness since hashtags can be used ironically or hijacked by opposing groups, making them an unreliable indicator of ideology. Beyond the specific limitations of each paper, we noticed a common gap across all four studies: none of them compared echo chamber behaviour across two distinct time periods within the same political event. Most studies analysed their data as a single snapshot, making it harder to say whether echo chambers become stronger or weaker as an election approaches.

Our study directly addresses these gaps. We use a dataset of tweets from the 2024 U.S. Presidential Election on X. Instead of using hashtags to label users politically, we take an approach where communities emerge naturally from positive mention interactions, and we only check their political identity afterwards by looking at the most influential accounts in each group. Most importantly, we analyse two separate time windows so we can directly compare echo chamber strength between the early campaign and the final weeks before election day. Our central research question is: *do users who positively mention each other on X during the 2024 U.S. Presidential Election tend to belong to the same ideological community, and does this tendency change as election day approaches?*

2 Methodology

2.1 Dataset

The data used in this project was sourced from a publicly available dataset of tweets related to the 2024 U.S. Presidential Election, collected and published by Balasubramanian et al. [1] from the University of Southern California. The dataset contains election-related tweets posted on X (formerly Twitter) between May and November 2024, covering the full campaign period leading up to election day. Each tweet includes fields such as the tweet text, author information, mentioned users, date and timestamp, engagement metrics (likes, retweets, views), hashtags, and language. For this project we worked with two temporal subsets of the

data: a May–July chunk representing the early campaign period, and an October chunk representing the final weeks before the election. These two windows were chosen to allow a direct comparison of network structure at two distinct stages of the electoral cycle, enabling us to assess whether echo chamber effects intensified as the election approached.

2.2 Data Preprocessing

Before building the networks, several preprocessing steps were applied to both time windows. First, author usernames and mentioned usernames were extracted from the raw user and mentionedUsers fields using regular expression parsing, since these fields were stored as nested JSON-like strings rather than plain text values.

Second, the tweet text was passed through the **VADER (Valence Aware Dictionary and sEntiment Reasoner)** sentiment analyser to compute a compound sentiment score for each tweet. VADER is a lexicon-based sentiment tool specifically designed for short social media text. It returns a score between 1 (most negative) and +1 (most positive), with scores above +0.05 considered positive, below -0.05 considered negative, and everything in between considered neutral. [10]

Third, we filtered out interactions where the sentiment compound score was below -0.05. The reasoning behind this is that negative interactions, such as criticizing or attacking another user, do not indicate ideological agreement. Since echo chambers are defined by users reinforcing each other's views, only neutral and positive interactions are meaningful for this analysis. Keeping negative mentions would introduce noise by connecting ideologically opposed users through conflict rather than agreement.

Fourth, we removed self-mentions (cases where a user mentioned themselves), as these carry no social interaction signal. Finally, we filtered the network to retain only the top 5,000 most active users by total mention frequency.

This step was taken to reduce computational overhead and to focus on the core of the political conversation.

2.3 Network Construction

For each time window, we constructed a directed weighted mention network. In this network, each node represents a unique user account, each directed edge from user A to user B indicates that user A mentioned user B in a tweet, and the weight of each edge represents how many times user A mentioned user B. The higher the weight, the stronger the connection between the two users. This representation captures active engagement between users, reflecting who is paying attention to whom within the political conversation. Two separate networks were built. One for the May–July period and one for October, to enable comparison.

2.4 Community Detection

To identify groups of users who predominantly interact with each other, we applied the **Louvain community detection algorithm** to each network. The Louvain method is an unsupervised algorithm that partitions a network into communities by maximising modularity — a measure of how densely connected nodes are within communities compared to what would be expected by chance. A higher modularity score indicates more clearly separated communities. [3]

Crucially, no political labels were provided to the algorithm at this stage. Communities were detected purely based on the structure of user interactions, without any prior knowledge of user ideology. This unsupervised approach ensures that any clustering found in the results emerges organically from the data rather than being imposed by us.

Because Louvain can return a large number of small communities, we retained only the top 10 largest communities by membership size for further analysis, discarding smaller peripheral clusters. It is important to also note that the graph was converted to undirected

graph because Louvain algorithm explicitly requires an undirected graph. [17]

2.5 Measuring Homophily via the E/I Index

To quantify the degree of echo chamber behaviour, we computed the **External/Internal (E/I) Index** for each detected community, following Krackhardt et al [12]. This metric measures the ratio of interactions that stay within a community (internal) versus those that cross into other communities (external). For each community, the index is defined as:

$$E/I = \frac{E - I}{E + I} \quad (1)$$

where E is the number of outward mentions directed at users outside the community, and I is the number of outward mentions directed at users inside the same community. The index ranges from -1 to $+1$, where:

- -1 indicates a perfect echo chamber: all interactions are internal to the community
- 0 indicates no preference: interactions are equally split between internal and external
- $+1$ indicates a fully outward-looking community: all interactions cross community boundaries

This was computed separately for each of the top 10 communities in both the May–July and October networks, allowing us to compare the strength of echo chamber behaviour across time periods and across individual communities. A community with an E/I index closer to -1 is interpreted as a stronger echo chamber, as its members predominantly interact among themselves rather than engaging with users from other political clusters.

3 Results

To answer our research question on whether users on X tend to interact more with people who share their political views, we analyse the interaction patterns in our mention networks using a set of measurements and visualisations.

We look at how the communities formed, which accounts were most influential within each community, and how isolated each community was from the others. The results below show the degree to which users moved towards the like-minded people, and form the basis for the discussion that follows.

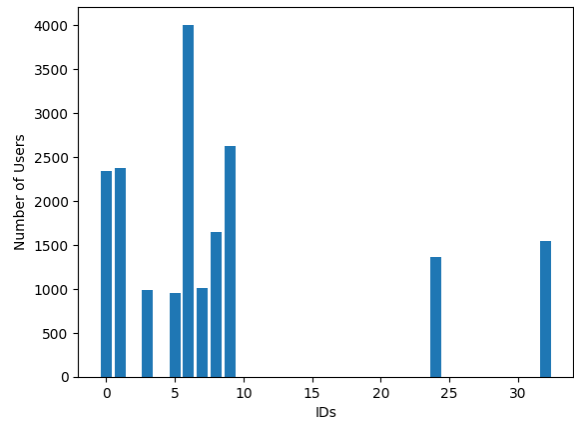


Figure 1: Top 10 Largest Communities From May

Figure 1 shows how large the top 10 detected communities in the network really were in May. Each bar corresponds to one community discovered by the Louvain algorithm. In this figure the X-axis represent the community ids and the Y-axis represents the number of users in each community.

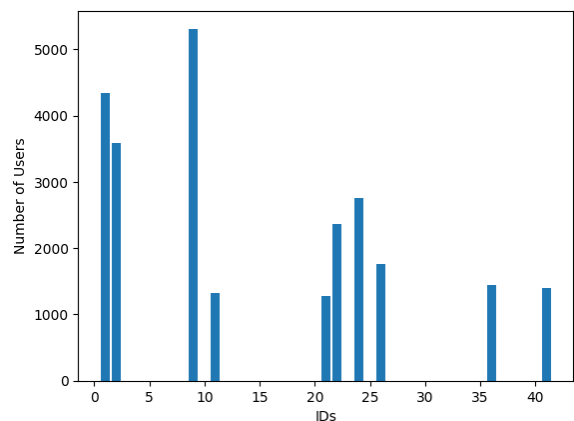


Figure 2: Top 10 Largest Communities From October

Figure 2 is the same as Figure 1 in how it is represented but Figure 2 now shows the top 10 largest detected communities in October. It clearly shows the top communities now are larger than in May.

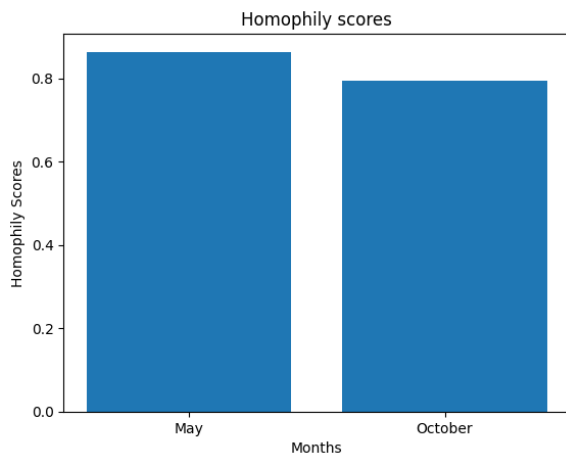


Figure 3: Homophily Score

Figure 3 shows the Homophily scores for May and October. It is clear that the homophily score was higher in May, but decreased only slightly in October. These numbers suggest that, at the start of the US election, more people were interacting with users who shared their political views than at the end.

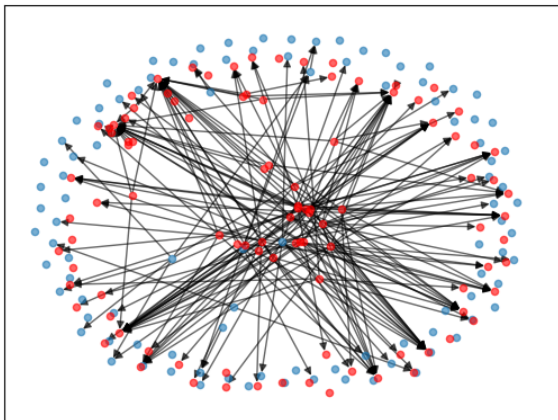


Figure 4

The visualization of Figure 4 shows the induced subgraphs of the 100 users with the most connections (as measured by degree centrality) in the networks from May to October. Each node represents a high-degree user, and the edges represent the interactions between them based on mentions. While the blue edges represent May, the red edges stand for October. The visualizations therefore focus on the “core” of the network and illustrate how the most influential users are connected to one another, rather than mapping the entire user population.

Figure 5 shows the ranked distribution of all

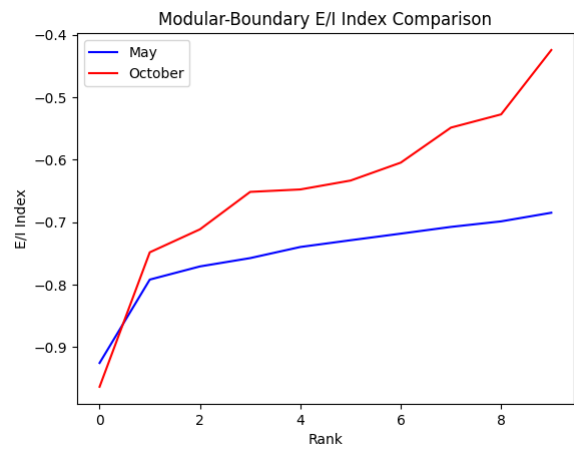


Figure 5

communities based on their External/Internal (E/I) index for May and October. Each point represents a community, ordered from those with the strongest internal connectivity (most echo chamber like structure) to the community with the strongest external connectivity. The graph allows for a comparison of how the openness or closedness of community boundaries differs between the two time periods.

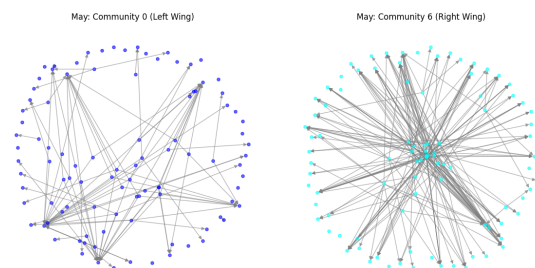


Figure 6

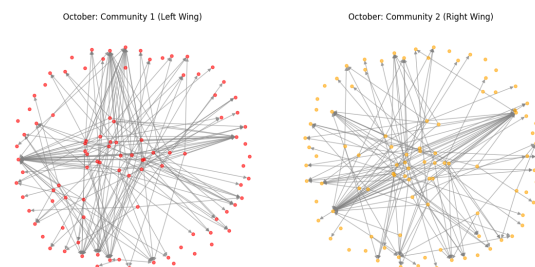


Figure 7

Figures 6 and 7 compare the interaction patterns of selected left-leaning and right-leaning communities, with Figure 6 representing May and Figure 7 representing October. For each community, the 100 users with the most

connections (measured by degree centrality) are extracted to form a subgraph. Nodes represent users, and edges denote interactions between them based on mentions. The resulting networks illustrate the internal connection patterns and structural differences within each ideological community.

(Note: Community IDs are assigned randomly by the Louvain algorithm and may differ across runs due to the nature of the algorithm. The community IDs reported here correspond to this specific run of the code and should not be treated as fixed identifiers.)

4 Discussion

The results of this study provide a clear evidence of echo chamber behaviour within the political Twitter/X discourse surrounding the 2024 U.S. Presidential Election. It reveals meaningful differences in the structure and intensity of these chambers between the early campaign period (May-July) and the weeks preceding election day (October).

4.1 Community Structure Reflects Real Political Divisions

One important finding is that the Louvain algorithm, operating purely on interaction structure without any political labels, successfully recovered communities that closely mirror real-world ideological divisions. In May, the largest community (ID 6, 4007 users) was dominated by @GOP, @realDonaldTrump, and @LaraLeaTrump, clearly identifying it as a right-wing cluster. By contrast, Community ID 0 (2345 users) centred around @harryjsisson and @mmpadellan - prominent left-wing commentators. In October, the structure became even cleaner: Community ID 1 (4343 users) consolidated around @KamalaHarris and @BarackObama, while Community ID 2 (3590 users) was dominated by @realDonaldTrump, @JDVance and @elonmusk. In May, we found 6 of the top 10 communities were right leaning with 3 of them left leaning. In October

however, 5 of the communities were right leaning, just 2 being left leaning, and the rest being either centre, news-based communities, or mixed/fringe communities.

4.2 Echo Chambers Are Strong Across Both Periods

All ten communities in both time windows returned strongly negative E/I indices, confirming the presence of echo chambers throughout the election period. In May, E/I scores ranged from -0.685 (Community 8) to -0.926 (Community 4), indicating that a large number of interactions remained within community boundaries. In October, scores ranged from -0.424 (Community 21) to -0.964 (Community 41). This is consistent with the findings of Flaxman et al. [8] and Cinelli et al. [5], who argue that ideological segregation and echo-chamber effects emerge from the properties of online information networks and persist across different contexts rather than being driven solely by specific political events.

4.3 Echo Chambers Weakened Slightly in October — With Notable Exceptions

Contrary to the expectation that polarization intensifies as election day approaches, the majority of communities showed a slight weakening of echo chamber behaviour in October compared to May. The large right-wing communities in October (IDs 2, 9, 21) posted E/I scores of -0.652, -0.633, and -0.424 respectively, which are less negative than the equivalent May right-wing clusters (ID 6: -0.771, ID 24: -0.926). Similarly, the main left-wing October cluster (ID 1: 0.648) is less insular than its May counterparts (ID 0: -0.758, ID 8: -0.685). One possible explanation is that as the election approached, users engaged more broadly across political lines, driven by increased media attention, cross-partisan criticism, and the increased visibility of the opposing candidate. This finding partially contradicts Leanne Kendall [11], whose work suggested that partisan engagement intensifies around political events.

4.4 The Notable Exception: Community 41

The one community that stood apart from all others was October Community ID 41 (0.964), the most insular community found across both time periods. Unlike the other clusters, its dominant account @wman132 (this account currently doesn't exist because it was suspended by X) was an account that held a centrality score of 0.0585, dramatically higher than any other account in the community. Meanwhile the remaining accounts had near-zero centrality scores. @idrissaberkane, the second most central account, is Idriss Aberkane, a French academic with three PhDs from École Polytechnique, whose content primarily focuses on politics. Nonetheless, this community should be treated as an anomalous cluster.

4.5 The Emergence of a Media Cluster in October

A notable structural difference between May and October is the appearance of Community ID 26 in October, centred around @FoxNews, @CBSNews, @MSNBC, and @CNN. This community, which has the second weakest echo chamber score of all October communities at -0.527, appears to represent a media and institutional cluster where users engage with both partisan and neutral news sources simultaneously. No equivalent community was found in May, suggesting that as the election drew closer, a segment of users gravitated toward cross partisan media consumption, which may have acted as a bridging layer between the two main ideological blocs.

4.6 Mixed and Fringe Communities

Several communities resist simple left/right classification. In May, Community ID 7 combined Trump family accounts with crypto-focused accounts (@WhaleFUD, @MagaHAT.ETH), suggesting an overlap between political identity and cryptocurrency culture — a phenomenon also observed by researchers studying the intersection of financial

populism and politics. Interestingly, this finding does not support the survey-based research by Littrell et al. [13], which argued that cryptocurrency owners are highly diverse and lean toward the left rather than a conservative side. However, our findings are highly supported by the social media text analysis of Banker et al. [2] who showed that cryptocurrency talk on X is fundamentally tied to conservative moral values rather than liberal ones.

Community ID 9 in May showed a genuinely mixed structure, with @JoeBiden and @POTUS appearing alongside right-wing accounts @LauraLoomer and @SteveLovesAmmo, likely reflecting the period of uncertainty around Biden's candidacy and the cross-partisan attention it attracted. These mixed communities are consistent with the argument by Enjolras et al. [6] that homophily and echo chambers do not always necessarily co-occur and that some user segments remain genuinely cross-partisan in their engagement patterns.

4.7 Implications

Overall, these findings suggest that while echo chambers are an important feature of U.S. political Twitter discourse, their intensity is not directly correlated with proximity to election day. Instead, the structure of political communities appears to be shaped by specific events and actors, such as Biden's withdrawal, the Harris nomination, and Elon Musk's increasing alignment with the Trump campaign in October. These may have reorganised how users clustered together and how openly they engaged across community boundaries.

5 Limitations

There are several limitations to keep in mind when reading these results. First, this study only uses data from X (formerly Twitter), a single social media platform. Echo chamber behaviour can look very different across platforms. For example, Cinelli et al. [5] performed a large-scale comparative analysis across Facebook, Twitter, Reddit, and Gab, finding

that echo chamber dynamics vary considerably depending on the platform’s structure and user base. Similarly, Luan et al. [15] studied opinion homophily across both Twitter and Reddit, highlighting that interaction patterns and community formation differ meaningfully between platforms. Since our analysis is limited to X and covers only English-language content, the findings may not generalise to other platforms or to political communities that primarily communicate in other languages.

Furthermore, the network model is limited exclusively to @mention interactions. Other key forms of user interaction, such as replies, quote tweets, or likes, are not taken into account due to increased complexity and high computational overhead we faced in regards to the network creation while we were coding. As a result, only part of the actual social dynamics is captured, which can lead to an incomplete representation of the interaction network. This is a notable limitation compared to prior work, for example, by Marques-Neto et al. [16], who constructed multiplex networks incorporating follow, mention, and retweet connections simultaneously, and found that homophily levels increased when multiple interaction types were combined. A multi-layer network approach of this kind would likely produce a more complete picture of ideological clustering, and represents a clear direction for future work.

Another limitation is related to the filtering of interactions based on the VADER sentiment score. By excluding negative interactions (scores lesser than 0.05), conflict-based forms of exchange that are potentially politically relevant are removed in particular. This can lead to bias. The measured homophily may be somewhat overestimated, since only neutral and positive interactions are retained, systematically excluding the cross-partisan engagement that would otherwise lower the assortativity scores. In addition, the ideological structure is indirectly derived through community detection using the Louvain algorithm. Although this method has been used multiple times earlier, the resulting cluster structure depends on algorithmic parameters

and network characteristics. Furthermore, converting the network into an undirected graph for community detection results in the loss of information about the direction of interactions, which can limit the interpretation of asymmetric communication patterns.

Moreover, the analysis is based solely on two discrete time periods (May(-July) and October), which means that short-term dynamics within the campaign are not captured. As a result, it remains unclear whether changes in homophily occur gradually or abruptly. Ultimately, the analysis was limited to the 5,000 most active users in the order to reduce computational complexity. However, this results in the exclusion of less active users, which may skew the results toward highly engaged political actors and potentially overemphasize the structure of the overall network.

Future work could address these limitations by incorporating additional forms of interactions, testing other social media platforms, comparing multiple community detection methods, and examining the echo chambers’ evolution over longer time periods.

6 Conclusion

This study tried to investigate whether echo chambers existed within the political discourse on X (formerly Twitter) during the 2024 U.S. Presidential Election, and whether their intensity changed over the course of the election campaign. Using a large-scale dataset of election-related tweets collected by Balasubramanian et al. [1], we constructed weighted mention networks for two windows (May–July 2024 (early campaign) and October 2024 (pre-election)) and applied the Louvain community detection algorithm in an unsupervised manner to identify emergent user clusters. Community validation via degree centrality confirmed that the detected clusters closely resembled real-world ideological divisions, with the largest communities in both periods dominated by either prominent right-wing figures such as @realDonaldTrump, @GOP, and @elonmusk, or left-wing figures such as

@KamalaHarris, @BarackObama, and @JoeBiden.

The E/I index scores computed for all ten communities in both time windows were strongly negative, ranging from -0.685 to -0.926 in May and -0.424 to -0.964 in October, confirming that echo chambers were a major feature of political Twitter discourse throughout the election period. Contrary to our initial expectation that echo chambers would intensify as election day approached, the majority of communities showed a slight weakening of modular-Boundary index in October compared to May. We interpret this as a higher level of cross-ideological engagement during the final weeks of the campaign, rather than a genuine reduction in ideological homophily.

Other additional findings were the emergence of a media cluster in October and a crypto-political community identified in May. This group (Community ID 7), combining Trump family accounts with cryptocurrency influencers, pointed to an intersection between financial populism and right-wing political identity. Community ID 41 in October, with an extreme E/I score of -0.948 represented the most insular cluster found across both periods, possibly as a potential fringe or coordinated behaviour cluster.

This study demonstrates that our approach successfully recovers ideologically coherent communities from mention interaction data alone, without requiring manual labelling. It also provided a comparison of echo chamber strengths across two stages of the 2024 U.S. Presidential Election, helping in analyzing evolution of echo chambers. It also applied the E/I index as a community-level echo chamber metric, offering a better view of insularity than other aggregate modularity scores used in most prior work.

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